



SYMBIOSIS INSTITUTE OF TECHNOLOGY
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Year: AY 25-29

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Semester: 3

ASSIGNMENT No: 5

TITLE: Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement :

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Theory:

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming

Program :

```
interface Vehicle
{
    void start();
}
class Car implements Vehicle
{
    public void start()
    {
        System.out.println("car is starting.... ");
    }
}
class ElectricCar extends Car
{
}
```

```

    void charge()
    {
        System.out.println("&quot;Electric Car is charging....&quot;");
    }
}

public class InheritanceInterfaceDemo
{
    public static void main(String[] args)
    {
        ElectricCar e = new ElectricCar();
        e.start();
        e.charge();
    }
}

```

Output :

```

C:\Users\CL407_11\Documents\java>javac InheritanceInterfaceDemo.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8

C:\Users\CL407_11\Documents\java>java InheritanceInterfaceDemo
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
car is starting....
Electric Car is charging....

C:\Users\CL407_11\Documents\java>

```

```

C:\Users\CL407_11\Documents\java>Date
The current date is: Sat 07/25/2026
Enter the new date: (mm-dd-yy)

```

Enter your choice: 1

===== TODAY'S LEARNING PLAN =====

1. Read about the discovery of nuclear fission.
2. Learn the difference between fission and fusion.
3. Study the strstructure of an atom.
4. Watch a documentary on nuclear energy.
5. Learn the peaceful uses of nuclear technology.
6. make nuclear bomb

===== LEARNING PLANNER =====

1. View Today's Learning Tasks

Exercise 1

Program :

```
abstract class Shape {  
  
    abstract void calculateArea();  
}  
  
class Circle extends Shape {  
  
    double radius;  
  
    Circle(double radius) {  
        this.radius = radius;  
    }  
  
    @Override  
    void calculateArea() {  
        double area = Math.PI * radius * radius;  
        System.out.println("Area of Circle: " + area);  
    }  
}  
  
class Rectangle extends Shape {  
  
    double length;  
    double width;  
  
    Rectangle(double length, double width) {  
        this.length = length;  
        this.width = width;  
    }  
  
    @Override  
    void calculateArea() {  
        double area = length * width;  
        System.out.println("Area of Rectangle: " + area);  
    }  
}
```

```

    }
}

public class ShapeApplication {

    public static void main(String[] args) {

        Shape circle = new Circle(5);
        Shape rectangle = new Rectangle(10, 5);

        circle.calculateArea();
        rectangle.calculateArea();
    }
}

```

Output:

```

1 error          javac ShapeApplication.j
avaC:\Users\HP\Documents\java>
PS C:\Users\HP\Documents\java> java ShapeApplication
Area of Circle: 78.53981633974483
Area of Rectangle: 50.0
PS C:\Users\HP\Documents\java>

```

2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

Source code:

```

import java.util.ArrayList;

interface Product {

    void displayProduct();
    double calculatePrice();
}

class ProductDetails {

    String name;
    double price;
}

```

```

    ProductDetails(String name, double price) {
        this.name = name;
        this.price = price;
    }
}

class Electronic extends ProductDetails implements Product {

    int warranty;

    Electronic(String name, double price, int warranty) {
        super(name, price);
        this.warranty = warranty;
    }

    public void displayProduct() {
        System.out.println("Product: " + name);
        System.out.println("Price: " + price);
        System.out.println("Warranty: " + warranty + " years");
    }

    public double calculatePrice() {
        return price;
    }
}

class Clothing extends ProductDetails implements Product {

    String size;

    Clothing(String name, double price, String size) {
        super(name, price);
        this.size = size;
    }

    public void displayProduct() {
        System.out.println("Product: " + name);
        System.out.println("Price: " + price);
        System.out.println("Size: " + size);
    }

    public double calculatePrice() {
        return price;
    }
}

```

```
}  
}
```

```
class Grocery extends ProductDetails implements Product {
```

```
    double weight;
```

```
    Grocery(String name, double price, double weight) {  
        super(name, price);  
        this.weight = weight;  
    }
```

```
    public void displayProduct() {  
        System.out.println("Product: " + name);  
        System.out.println("Price: " + price);  
        System.out.println("Weight: " + weight + " kg");  
    }
```

```
    public double calculatePrice() {  
        return price;  
    }  
}
```

```
public class ECommerceSystem {
```

```
    public static void main(String[] args) {
```

```
        ArrayList<Product> products = new ArrayList<>();
```

```
        products.add(new Electronic("Laptop", 60000, 2));  
        products.add(new Clothing("T-Shirt", 999, "L"));  
        products.add(new Grocery("Rice", 1200, 10));
```

```
        for (Product p : products) {
```

```
            p.displayProduct();  
            System.out.println("Final Price: " + p.calculatePrice());  
            System.out.println("-----");
```

```
        }  
    }  
}
```

Output:

```
PS C:\Users\HP\Documents\java> java ECommerceSystem
-----
Product: T-Shirt
Price: 999.0
Size: L
Final Price: 999.0
-----
Product: Rice
Price: 1200.0
Weight: 10.0 kg
Final Price: 1200.0
-----
❖ PS C:\Users\HP\Documents\java> 
```